



ELECT MAINT- OFF SITE Department
SIMHADRI



Ref:SMTTP/ELECT MAINT-/2023-24/LMI/126518

Date:

16/05/2023(16:37:59)

Subject:SCHEDULE REVIEW OF LMI-STATION PREPAREDNESS FOR
LIGHTNING & THUNDERSTORM

Subject:LMI-STATION PREPAREDNESS FOR LIGHTNING & THUNDERSTORM

LMI/ OGN/OPS/SYST/031 on Station Preparedness for Lightning & Thunderstorm is reviewed as per schedule and

no changes are envisaged.

Review proposal of LMI is submitted for the kind approval.

Approving authority for above proposal is HEAD OF THE PROJECT

Submitted by:

जे एस एन वी प्रसाद - वरिष्ठ प्रबंधक

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On:16/05/2023(16:37:59) **Ref:**SMTTP/ELECT MAINT-/2023-24/LMI/126518

No changes in the document except periodic review.

May be Approved.

Submitted by:

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Concern department may please enclose the check list as per LMI/OGN of STATION PREPAREDNESS FOR LIGHTING & THUNDER STROM.

Submitted by:

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Pl review

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LMI on Station preparedness for Lightning and Thunderstorm reviewed and put up for further approval.

Submitted by:

जी एच राव - अपर महाप्रबंधक

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Reviewed and forwarded for further review/approval

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May pl see

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Checked the LMI. May Pl forward.

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On:22/05/2023(10:29:27) **Ref:**SMTTP/ELECT MAINT-/2023-24/LMI/126518

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Mentioned roles of Offsite & Civil are mentioned/included in LMI/ OGN/OPS/SYST/030 on Station Preparedness for Cyclone.

Submitted by:

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review

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LMI reviewed.

Forwarded for further approval.

Submitted by:

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Submitted for approval

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Under Page No 7 (Srl (B) (1) - Sending SMS Regarding Likely Time & Duration of Storm Regg.

Bulk SMS Now is Allowed only Confirming to Approved Patterns. Hence Open Ended SMS Cannot be Sent.

The Following Modification may be Done.

WhatsApp Group to be Formed for Sending Information. Further The Provider of Information can Directly Send the Information to the Intended Consumers. Suggested Dept for Such Activity would be EMG Dept.

Note Initiator may Review & Do the Needful

Submitted by:

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On:01/06/2023(10:36:22) **Ref:**SMTTPP/ELECT MAINT-/2023-24/LMI/126518

R/Sir

May please see the notings of AGM (IT) in the notesheet No.9

Submitted by:

जे एस एन वी प्रसाद - वरिष्ठ प्रबंधक

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On:03/06/2023(18:02:02) **Ref:**SMTTPP/ELECT MAINT-/2023-24/LMI/126518

Already **Operation interface** whatsapp group is available for sending such information. No seperate whatsapp group is required please.

Submitted by:

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On:03/06/2023(19:29:44) **Ref:**SMTTPP/ELECT MAINT-/2023-24/LMI/126518

For necessary review please.

Submitted by:

जे एस एन वी प्रसाद - वरिष्ठ प्रबंधक

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On:06/06/2023(11:11:45) **Ref:**SMTTPP/ELECT MAINT-/2023-24/LMI/126518

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forwarded

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Submitted by:

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May review please

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Submitted by:

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May please approve

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NTPC Limited

SIMHADRI SUPER THERMAL POWER PROJECT

LOCATION MANAGEMENT INSTRUCTION

LMI/ OGN/OPS/SYST/031

Rev.0, May 2023

**STATION PREPAREDNESS FOR LIGHTNING &
THUNDERSTORM**

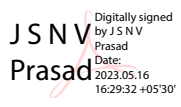
APPROVED FOR

IMPLEMENTATION BY - - - - -

HEAD OF THE PROJECT (SIMHADRI)

Date: _____

LMI APPROVAL DOCUMENT

1	Title of LMI	Station Preparedness for Lightning & Thunderstorm
2	a) Document Code b) Revision/Issue No. & Date	LMI/ OGN/OPS/SYST/031 Rev.0 May 2023
3	Original/Ref. document	COS-ISO-00-OGN/OPS/SYST/031
4	Prepared by	 JSNV Prasad Date: 2023.05.16 16:29:32 +05'30' SENIOR MANAGER (EM)
5	Checked by	AGM (EM)
6	Reviewed by	Design : HOD- (EM) Design : HOD- (C&I) Design : HOD- (EEMG) Design : HOD- (CHEM) Design : HOD- (CIVIL & OFS) Design : HOD- (TMD) Design : HOD- (BMD) Design : HOD- (SAFETY)

		Design : HOD- (IT) Design : HO-HR Design : CMO Design : AGM (OPN) Design : GM (FM) Design : GM (Maint) Design : GM (O & M)
7	Approved for issue	Signature : Design : GGM (Simhadri)
8	Date of Issue	
9	Index List	Enclosed
10	Distribution List	Enclosed

STATION PREPAREDNESS FOR LIGHTNING & THUNDERSTORM

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Station Preparedness for Lightning & Thunderstorm

1.0 Introduction

A thunderstorm, also known as a lightning storm, or thundershower, is a storm characterized by the presence of lightning and its acoustic effect on the Earth's atmosphere. Thunderstorms are usually accompanied by strong winds, heavy rain, and sometimes hail, or, in contrast, no precipitation at all. While severe thunderstorms are most common in the rainy season, they can occur at just about any time of the year.

Thunderstorms can pose great hazards. Damage that results from thunderstorms is mainly inflicted by downburst winds, large hailstones, and heavy precipitation.

In past, numbers of thunderstorms have resulted in loss of machine availability and property. Based on learning during thunderstorms, it is necessary that a set of guidelines be prepared to tackle thunderstorm related issues and emergencies. This will help in maintaining equipment availability & ensuring system revival in the event of outage in minimum possible time.

2.0 Objective and Scope

The objective of this LMI is to ensure station preparedness to effectively deal with thunderstorm.

These guidelines recognize that thunderstorms can be tackled with better preparedness to minimize losses and for faster stabilization of units, in case of tripping. This will also help in reducing system revival time in case of emergency situations like power supply failure, non-availability of DG set etc.

3.0 Superseded Documents Nil.

4.0 Responsibility

Based on these guidelines all the stations were asked to prepare LMI to ensure thunderstorm preparedness at the earliest. For smooth handling of thunderstorms and related information flow, following guidelines are to be followed:

- As per the OPERATION GUIDANCE NOTE: COS-ISO-00-OGN/OPS/SYST/030 it has advised to create a station level Cyclone Management Group (CMG). The same group shall take care of Management of Thunderstorm also.
- Head of O&M will be the coordinator/nodal officer for management of Cyclone as well as Thunderstorm also.
- Apart from O&M members, HR, Medical, CISF and canteen representative should also be the member of this team.
- Coordinator/ Nodal officer will be responsible for all information flow to RHQ and Corporate office

- This group will start working after receiving first intimation about expected thunderstorm.
- Head of Operation at COS will be the nodal point to give primary information report about the crisis to ED (OS) / Director (O).
- Station Coordinator/Nodal officer, concerned GM-OS shall be responsible for all the thunderstorm related activities of the station.

5.0 Thunderstorm preparedness

Most thunderstorms come and go fairly uneventfully. However, any thunderstorm can become severe, and all thunderstorms, by definition, present the danger of lightning. Therefore, thunderstorm preparedness refers to taking steps before, during, and after a thunderstorm to minimize injury and damage.

A) Routine activities to avoid disaster during Thunderstorm

Sl No.	Activity	Details f activity	Responsibility.
1.0	Checking of Lightning arresters of buildings, F.O. tanks etc.	All the high rise, critical buildings and chimney, F.O. tanks etc. must be checked regularly for healthiness of Lightning arresters installed. Earth resistance should be checked regularly & readings be recorded in line with details given in LMI/OGN/OPS/ELEC/021	Electrical maint. Dept.
2.0	Checking of Switchyard Lightning Masts, Lighting towers, earth wires etc.	Structures of Switchyard Gantry, Lightning Masts, Lighting towers, earth wires etc. must be checked for integrity as these are most vulnerable to create station contingencies.	Electrical maint. Dept.
3.0	Removal of Loose metallic sheets, trees, branches, tarpaulin sheets etc.	Remove all metallic sheets which could cause disaster to Electrical equipment nearby. Remove dead or rotting trees and branches that could fall and cause injury or damage during a severe thunderstorm. Secure outdoor objects that could blow away or cause damage.	All concerned Depts.
4.0	Clear station drainage system	All peripheral/ storm water drains, switchyard drainage system etc. shall be maintained to have proper flow of	Civil Maint. Dept.

		rain water	
5.0	Availability of normal/ emergency dewatering pumps	All dewatering pumps at critical locations should be available to meet contingency.	Mech./Offsite/ Civil/ Opn./CHP Maint. Dept.
6.0	Checking of level switches of Drain pits	Ensure that Level Switches of all drain pits are working properly for auto start of dewatering pumps.	C&I dept.
7.0	Availability of DC/ Emergency Lighting	Regular checking of auto operation of DC lighting at all locations must be ensured.	Operation/ Electrical Maint. Dept.

B) Readiness before Thunderstorm & Lightning

Sl No.	Activity	Details f activity	Responsibility.
1.0	Sending SMS regarding likely time & duration of Thunderstorm	Reliability of Service provider for mobile connectivity may be ensured. To all Cyclone/ Thunderstorm Management Group members. Announcement to all employees to suspend / postpone outdoor works & to remain indoors to avoid injury due to lightning, falling sheets, glass pieces etc during Thunderstorm. To close tightly all doors/ windows in all offices/ buildings.	IT Dept. Cyclone/ Thunderstorm Management Group members.
2.0	Communication within plant premises & township	Satellite phones (if available) to be made ready. Walkie-Talkies to be made ready with battery chargers. Backup Power Supply from DG Set to telephone exchange.	All concerned Depts.
3.0	Healthiness of Disturbance/ Event Recorder	Availability of all these equipment & its synchronization with main plant system to be ensured.	Elect./ C&I Dept.
4.0	Readiness of all DG	Trial run to be taken prior to Thunderstorm. All Diesel Tanks to be	Operation

	sets	made full	dept.
5.0	Readiness of Battery operated equipment (Trolley, Search lights, UPS etc.)	Batteries to be in fully charged condition. Float/ Boost Charging of all batteries to be ensured.	All concerned Depts.
6.0	Dewatering Pumps	All Dewatering Pumps / Diesel Dewatering Pumps at critical locations e.g. CW Pump Houses, CHP, TG Hall areas etc. to be kept in Ready-to-start mode. Similarly, Surface drainage pumps, GT Yard drainage pumps should be kept ready as above.	All concerned Depts.
7.0	Rain Coats, Gum Boots, Torches	Availability/readiness of these items at all critical locations may be ensured.	All concerned Depts.
8.0	Emergency Lights / search lights	Emergency Lights to be checked physically & made ready at all locations. Additional torches along with cells can be made available at all operational locations and to maintenance personnel. Search lights to be made available.	All concerned Depts.
9.0	Availability of Condenser diaphragms	Condenser diaphragms may be kept ready for all units	Turbine Maint. Dept.
10.0	Switchyard Control Room & Main Unit Control Room	A group of Sr. executives from Operation, Electrical, C&I and Turbine maint. to remain present in Main Plant control room. Similarly, Sr. executives from Operation & Switchyard maint. Group to remain present at Switchyard Control Room to handle any emergency situation. Groups to work in close association with SCE. No Work Permits shall be planned for EHV System or major auxiliaries. SCE to remain in touch with Grid officials. All MCCs Doors and windows to be ensured for tight closure.	All concerned Depts

11.0	Bunker Level	Bunker levels in running units to be made full. Conveyors to be stopped before Thunderstorm. All persons to be asked to come out of conveyor gallery, Track hopper areas.	Main plant, CHP Operation & CHP Maint.
12.0	Coal Handling Plant	Stacker Reclaimer should be parked with its boom properly anchored.	CHP Operation & Maint
13.0	Track Hopper	Track Hoppers should be evacuated fully to avoid overflowing of Slurry Coal	CHP Operation & Maint.
14.0	MGR System	Manual brakes should be applied to the loose Railways/ MGR wagons to avoid rolling down of wagons	MGR Operation & Maint.
15.0	Shift Operations	Station should be ready to quickly reduce load as per the prevailing grid frequency. As the reduced demand during storms is of short duration, Generating station must pick up load immediately as per grid frequency when load starts getting connected to the grid. All these may be done under intimation to respective RLDCs	Operation Dept. & EMD
16.0	Parking of Lifts	All lift to be parked at zero meter. Lift Pit dewatering facility to be kept ready.	EMD
17.0	Medical Aid	Medical Team along with Ambulance to be on duty round-the-clock at Main Plant	Medical Cell
18.0	Fire & Safety	Fire and Safety Personnel to be available round-the-clock at Main Plant.	Safety Dept./ CISF
19.0	Availability of food stuff, Drinking water	Adequate stock of snacks/ food material/ Drinking water at Control rooms & different manned locations to be arranged.	HR dept.

20.0	Ambulance preparedness	Ambulance at safe plant location to urgently take care of casualties.	SCE & Medical & Safety.
21.0	Safety of personnel	No manpower movement at outdoor locations during Thunderstorm. No person shall stand near trees / tall masts. A team should be ready for removal of road & MGR track blockage caused by possible uprooting of trees and poles.	All Concerned Dept.
22.0	EOT/ Gantry Cranes	Locking of EOT/ Gantry Cranes must be ensured	Mech. Maint.
23.0	Hydra, JCB, Portable DG Set, Power Saw	Availability/readiness of these equipment along with operators may be ensured.	All concerned Depts.
24.0	Big trees/ poles/ towers on approach road from Township to Plant & within plant premises	Identify all big trees/ poles/ towers which may uproot due to high velocity winds during thunderstorm & take precautionary measures.	Civil maint.
25.0			

C) During Thunderstorm & Lightning

Sl No.	Activity	Details f activity	Responsibility.
1.0	Stay informed	To remain in touch with IMD, news sources, weather information centre etc. for latest updates & conveying the same thro' SMS	Cyclone/ Thunderstorm Nodal Group members
2.0	Safety of personnel	Avoid contact with corded phones and devices including those plugged into electric for recharging. Cordless and wireless phones not connected to wall outlets are OK to use. Avoid contact with plumbing. Avoid washing hands, dishes, taking a shower as plumbing	All concern.

		and bathroom fixtures can conduct electricity during lightning. Do not touch materials or surfaces that can conduct electricity, including metal scaffolding, metal equipment, utility lines, water, water pipes and plumbing. Stay away from windows, doors and porches. Take shelter in a sturdy building. Avoid isolated sheds or other small structures in open areas.	
3.0	Check for water ingress	Check for water ingress in control rooms/ on Electrical & Electronic panels and take immediate action to prevent it.	Elect./ C&I Dept.
4.0	Treatment of lightning victim	Follow these steps if someone has been struck by lightning: Call for Emergency Medical help. Anyone who has sustained a lightning strike requires professional medical care. Check the person for burns and other injuries. If the person is breathing normally, look for other possible injuries and care for them as necessary. People who have been struck by lightning do not retain an electrical charge and can be handled safely.	All concern.

D) After a Thunderstorm & Lightning

Sl No.	Activity	Details of activity	Responsibility.
1.0	Updation of information	Listen to the news sources about the improvement in situation & communicate to all.	Cyclone/Thunderstorm Nodal Group
2.0	Assessment of the situation	Survey the outside and inside of all the plant buildings, EOT/ Gantry cranes, Chimney, FO Storage tanks, Ash	All concerned Depts.

		Dyke, outdoor Equipment, like, Boilers, Switchyard, transmission towers, Transformers, OH traction lines etc. for structural damage. Look for sharp objects, downed power lines, damaged pipe lines, or other hazards. Intimate concerned department for quick restoration/ repair	
3.0	Clean up safely	All the debris, flying objects, unwanted material, and access to blocked roads may be cleared quickly. Persons involved in this exercise must wear proper PPE before taking up the job.	All concerned Depts.

6.0 Format for information flow to Corporate Centre

6.1 Format for MIS on expected Thunderstorm

Name of the station :- NTPC-Simhadri

Sl. no.	Expected date/time of occurrence	Expected area affected

6.2 Format for MIS on Thunderstorm

Name of the station:- NTPC- Simhadri

Date and time of the thunderstorm: -

Sl. no.	Duration of thunderstorm	Area affected	Extent of damage	Expected normalization time.

7.0 Suggested enablers for effective handling of Thunderstorms

- Availability of details on alternate source of power in case of grid failure, start-up power scheme
- Availability of latest grid normalization procedure.
- Availability of contact nos. of all identified executives at station, RHQs, CC, RLDCs and other important persons.

- Availability of web addresses of concerned organizations <http://www.imd.gov.in/>,
<http://www.rsmcnewdelhi.imd.gov.in/>

8.0 REVIEW

This LMI document will be reviewed once in three years by Head of the project, NTPC Simhadri.

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REVISION INDEX

(For Changes / Revision)

Sl.No.	Rev. No.	Date	Details	Approved By
1	Original	May'2021	I st Issue	GGM (Simhadri)
2	Rev 0	May'2023	I st Issue	GGM (Simhadri)